



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

B.Sc. PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	CHARACTERIZATIONS AND APPLICATIONS OF NANOMATERIALS (GROUP I: TECHNIQUES IN MATERIALS PHYSICS)				
Type of Course	Vocational Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	1. PHY1VN101- Introductory Materials Science 2. PHY3VN201- Characterizations and Applications of Nanomaterials				
Course Summary	Master the art of characterizing nanomaterials with microscopy, diffraction, and spectroscopy techniques, unlocking their secrets and exploring their diverse applications.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO 1	Describe various microscopic techniques: Understand the principles and applications of optical, confocal, SEM, TEM, SPM, STM, and AFM for nanomaterial characterization.	U	F	Instructor-created exams / Quiz
CO 2	Explain diffraction principles: Grasp Bragg's law, crystal structure factors,	Ap	C	Instructor-created exams /

	and how X-ray diffraction (XRD) reveals nanomaterial structure.			Home Assignments
CO 3	Select appropriate characterization technique: Analyze nanomaterial properties and choose suitable techniques for specific information needs.	An	P	Seminar Presentation / Group Tutorial Work
CO 4	Operate characterization instruments: Gain practical experience using microscopy, diffraction, and spectroscopy tools for data acquisition.	E	P	Practical Assignment / Observation of Practical Skills
CO 5	Interpret characterization data: Analyze data from different techniques to extract information about size, morphology, structure, and composition.	E	P	Practical Assignment / Observation of Practical Skills
CO 6	Communicate characterization results: Effectively present findings using figures, graphs, and scientific language in written and oral formats.	E	P	Seminar Presentation / Writing Assignment
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Microscopic Techniques		9	15
	1	Optical microscope, Confocal microscope	2	
	2	Scanning Electron Microscopy (SEM)	2	
	3	Transmission Electron Microscopy (TEM)	2	
	4	Scanning Probe Microscopy (SPM), Scanning Tunneling Microscopy (STM), Atomic Force Microscopy (AFM)	3	
	Section 7.1 to 7.4, Book 1			
II	Diffraction Techniques		9	15
	5	X-ray Diffraction (XRD), Atomic scattering factor	2	
	6	Bragg's law	2	
	7	Crystal Structure factor, Diffraction from nanoparticles	2	

	8	X-ray diffractometer	2	
	9	Dynamic Light Scattering (DLS).	1	
	Section 7.5.1 to 7.5.8, Book 1			
III	Spectroscopic Methods		12	25
	10	Optical spectroscopy	2	
	11	UV-Visible spectroscopy	1	
	12	Infrared spectroscopy, FT-IR spectroscopy	3	
	13	Raman Spectroscopy	2	
	14	Photoluminescence Spectroscopy	1	
	15	X-Ray and Ultra Violet Photoelectron Spectroscopies	3	
	Section 7.6.1 to 7.6.8, Book 1			
IV	Applications of Nanomaterials		15	15
	16	Nanofluids, Hydrogen Storage	2	
	17	Solar Energy, Antibacterial Coating	3	
	18	Giant Magnetoresistance	2	
	19	Single electron Transistor	2	
	20	Self cleaning coating, nanotextiles, biomedical applications	2	
	21	Nanopore filters, water treatment	2	
	22	Nanodiamond, catalysts	2	
	Section 14.1 to 14.15, Book 2			
V	PRACTICALS		30	
	1	Characterization of prepared samples by any one of the techniques discussed in the syllabus		
Books and References: 1) Nanotechnology_ Principles and Practices by Sulabha K. Kulkarni (Book 1) 2) Nanomaterials and Nanocomposites: Synthesis, Properties, Characterization Techniques, and Applications by Rajendra Kumar Goyal (Book 2) 3) Springer Handbook of Nanomaterials by Robert Vajtai				

Mapping of COs with PSOs and POs :

	PS O1	PS O2	PSO 3	PSO 4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	1	1	0	2	1	1	1	1	1	2	1	1
CO 2	2	1	2	0	2	1	1	1	1	1	2	1	1
CO 3	2	1	3	1	2	1	1	1	1	1	3	1	1
CO 4	2	2	1	1	1	2	1	1	1	1	2	1	1
CO 5	2	2	2	0	2	1	1	1	1	1	2	1	1
CO 6	2	1	1	0	1	2	1	2	1	1	3	1	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignm ent	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	